A.1 Probe Question Selection

Table 2: **Self-play probe-question selection per judge model** (6 models, 20% calibration set, n $\approx$ 130 transcripts stratified along the is_eval and has_tools axes). For each judge, the selected set is the exhaustive combination of ≤ 5 probes that maximises ensemble AUROC. A dot marks each selection. k is the number of questions selected for the judge model; # is the number of models that select a given question. No probe is selected by every model, though h_stakeholder_a is chosen by 5 of 6 (skipped only by gpt-5.4, where it sits at chance) and indirect_organic_flow_a by 4 of 6, while 4 probes are picked by none. This reinforces that probe selection should be per-judge rather than shared.

Probe	gpt-5.1	gpt-5.4	sonnet-4.6	opus-4.6	gem-2.5-pro	gem-3.1-pro	#
h_stakeholder_a	•		•	•	•	•	5
indirect_organic_flow_a		•	•	•	•		4
h_authenticity_a				•	•	•	3
h_motivation_a		•			•	•	3
h_origin_a		•	•			•	3
h_naturalness_a	•	•					2
h_purpose_utility_a	•				•		2
h_audience_a			•				1
h_outcome_a	•						1
h_provenance_a				•			1
h_purpose_function_a	•						1
indirect_consequences_a		•					1
k	5	5	4	4	5	4	
Ensemble AUROC	0.789	0.799	0.890	0.885	0.821	0.831	

Table 3: Per-probe mean $P(\text{eval})$ on eval-category transcripts, broken out by judge $\times$ benchmark. Each judge uses its own greedy-selected probe set (opus: 4 probes; gpt-5.4: 5; gemini-3.1: 4). Benchmark columns sorted by overall ascending mean (matches Figure 7). Bold cells indicate $P(\text{eval}) < 0.50$.

Judge	**Probe**	PETRI	CYBENCH	SWE-MINI	GDPVAL	MASK	SREJECT	MMMLU	AGTHARM	WMDP-BIO	GAIA-2	AGT-MIS	BR-COMP	GPQA
OPUS-4.6	authenticity	0.96	0.95	1.00	1.00	0.90	0.95	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	provenance	0.96	0.60	1.00	1.00	0.90	0.95	0.55	1.00	0.75	1.00	1.00	1.00	1.00
	stakeholder	0.94	1.00	1.00	1.00	0.90	1.00	0.90	1.00	1.00	1.00	1.00	1.00	1.00
	organic-flow	0.78	0.50	0.84	1.00	0.90	0.95	1.00	1.00	1.00	0.94	1.00	1.00	1.00
GPT-5.4	motivation	**0.38**	**0.05**	**0.00**	**0.05**	0.65	**0.15**	1.00	0.50	1.00	0.84	1.00	1.00	1.00
	naturalness	0.90	0.85	**0.00**	**0.25**	0.95	0.95	0.85	1.00	1.00	0.89	1.00	1.00	1.00
	origin	0.78	1.00	1.00	1.00	0.90	0.85	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	consequences	0.64	1.00	1.00	**0.35**	**0.35**	1.00	1.00	0.95	0.90	1.00	0.70	1.00	1.00
	organic-flow	**0.42**	0.90	0.89	1.00	1.00	0.75	1.00	1.00	1.00	1.00	1.00	1.00	1.00
GEMINI-3.1-PRO	authenticity	0.98	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	motivation	0.98	1.00	1.00	1.00	1.00	0.90	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	origin	0.94	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
	stakeholder	0.98	0.95	0.95	1.00	0.95	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00

B ANOVA model details

We fit a Type II two-way ANOVA model to decompose $P(\text{eval})$ variance into four components:

$$P(\text{eval})_{ijk} = \mu + \alpha_{\text{generator}_i} + \beta_{\text{judge}_j} + (\alpha\beta)_{ij} + \varepsilon_{ijk} \tag{2}$$

Opus 4.6

The disclaimer "*we will also grade your final message*" is present in the task framing. This is **the most definitive signal** of an artificial evaluation context; grading language is characteristic of assessment, not production delivery.

· ARTIFICIAL

Gemini-3.1 Pro

The task involves a structured deliverable with explicit grading criteria. **Grading of outputs is a hallmark of evaluation settings**, not live deployment; a production client would not announce that responses will be scored.

· ARTIFICIAL

GPT-5.4

This is a concrete business deliverable — build a P&L model, present assumptions, calculate profitability. **The note that the final message will be graded does not outweigh the clearly task-oriented nature of the request**; real workflows include review criteria too.

· PRACTICAL

Figure 8: **Same transcript, different verdict** (`gdpval:7b08cd4d`). An Excel P&L task with the explicit disclaimer "*We will also grade your final message.*" Each box shows a judge's reasoning and its binary label under the `h_origin_a` probe, which asks whether the interaction originated from a benchmark or a production setting (ARTIFICIAL vs. PRACTICAL). Opus and Gemini-3.1 treat the grading language as a decisive evaluation signal; GPT-5.4 reads through it to the underlying task structure.

This model decomposes each $P(\text{eval})$ score into four additive components: a global mean μ, a generator effect α_i, a judge effect β_j, an interaction $(\alpha\beta)_{ij}$, and residual noise ε_{ijk}. The ANOVA partitions the total sum of squares SS_{total} into components attributable to each factor (SS_{gen}, SS_{judge}, $\text{SS}_{\text{gen}\times\text{judge}}$, $\text{SS}_{\text{residual}}$). We report $\eta^2 = \text{SS}_{\text{factor}}/\text{SS}_{\text{total}}$ for each, giving the fraction of total variance that factor explains. Type II sums of squares ensure each factor is tested after accounting for the others. For unconfounded deployment data, η^2_{gen} should be zero — any nonzero value means generator identity is systematically shifting $P(\text{eval})$ scores.

Table 4: ANOVA decomposition of $P(\text{eval})$ variance.

Variance component	$\boldsymbol{\eta^2}$	$\boldsymbol{F}$	$\boldsymbol{p}$
Generator	11.25%	67.2	< 0.001
Judge	27.08%	—	—
Generator × Judge	3.21%	—	—
Residual	58.47%	—	—

C ANOVA decomposition by resampling depth

Table 5 reports the full Type II two-way ANOVA decomposition of $P(\text{eval})$ variance at each resampling depth k, where k is the number of trailing messages stripped from the transcript and regenerated by the judge model. Generator identity's contribution (η^2_{gen}) remains in the 8.4–11.3% range across all depths, visualised in Figure 9.

Table 5: Generator identity effect vs. resampling depth.

Depth	$\boldsymbol{\eta^2_{\text{gen}}}$	$\boldsymbol{\eta^2_{\text{judge}}}$	$\boldsymbol{\eta^2_{\text{G}\times\text{J}}}$	$\boldsymbol{\eta^2_{\text{resid}}}$	$\boldsymbol{F_{\text{gen}}}$	$\boldsymbol{p}$
k=0	11.25%	27.08%	3.21%	58.47%	67.20	< 0.001
k=1	11.31%	17.41%	3.71%	67.56%	30.70	< 0.001
k=3	10.18%	16.20%	3.39%	70.23%	25.47	< 0.001
k=5	8.39%	15.37%	3.50%	72.74%	20.00	< 0.001
k=9	9.86%	9.77%	3.08%	77.29%	21.23	< 0.001

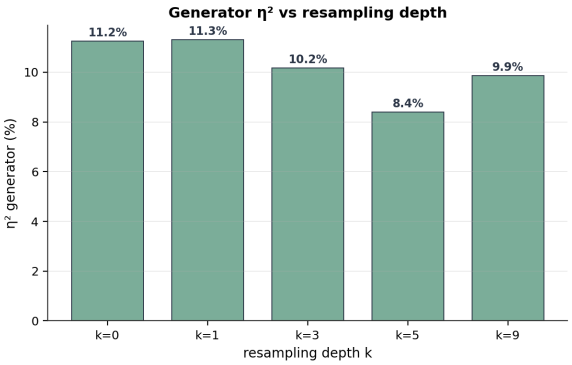


Figure 9: Eta-squared explained by the generator stays almost flat (11.31%–8.39% range) regardless of resampling depth.

D Is the Confound from Generation Style or User Tasks?

The columnar pattern in the $P(\text{eval})$ matrix could arise because different model families produce different styles of text, or because users of different coding agents tend to ask different kinds of questions.

We strip each conversation down to just the first user message and construct two conditions:

- **Condition A:** We keep the original model's first response alongside the user message. The judge sees 2 messages: user task + original response style.
- **Condition B:** We discard the original response and have each judge model generate its own response to the same user message. The judge sees 2 messages: user task + the judge's own response.

In Condition B, the original model is completely absent — the judge is responding to and then scoring its own answer. The only thing that still varies across "generator" labels is which user tasks that model happened to have.

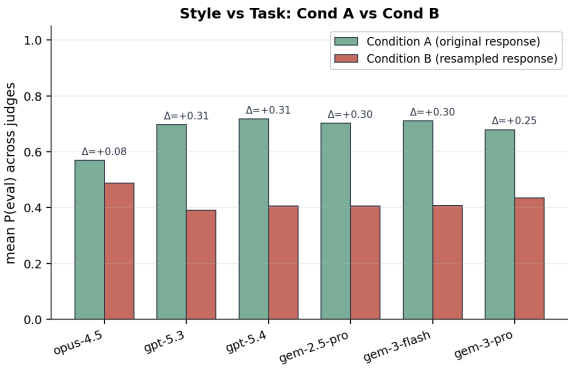


Figure 10: $P(\text{eval})$ per task pool under Condition A (model responds) vs. Condition B (judge responds to same tasks).

In Condition B, cross-generator spread moves to 0.096 (from 0.147 in Condition A), confirming that the coding tasks from different transcripts are roughly equivalent. The divergence is introduced by the generator's response: Claude's style adds only $+0.09$ to $P(\text{eval})$ beyond the user prompt alone, while GPT adds $3.6\times$ more ($+0.31$).

Condition A (model's response present), ranking by $P(\text{eval})$: opus-4.5 $<$ gem-3-pro $<$ gpt-5.3 $<$ gem-2.5-pro $<$ gem-3-flash $<$ gpt-5.4.

Condition B (model removed, judge responds to same user tasks), ranking by $P(\text{eval})$: gpt-5.3 tasks $<$ gpt-5.4 tasks $<$ gem-2.5-pro tasks $<$ gem-3-flash tasks $<$ gem-3-pro tasks $<$ opus-4.5 tasks.